Agentic Forecasting Bootcamp

The Learn Days — what to expect

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Vector Institute | July 2026

Why forecasting?

A uniquely rich testbed for agentic AI

Unstructured signals

Reason over news, policy, and reports — not just numbers.

Objective truth

Every prediction is scored against what actually happened.

Learns from experience

Expertise accumulates over many forecasting episodes.

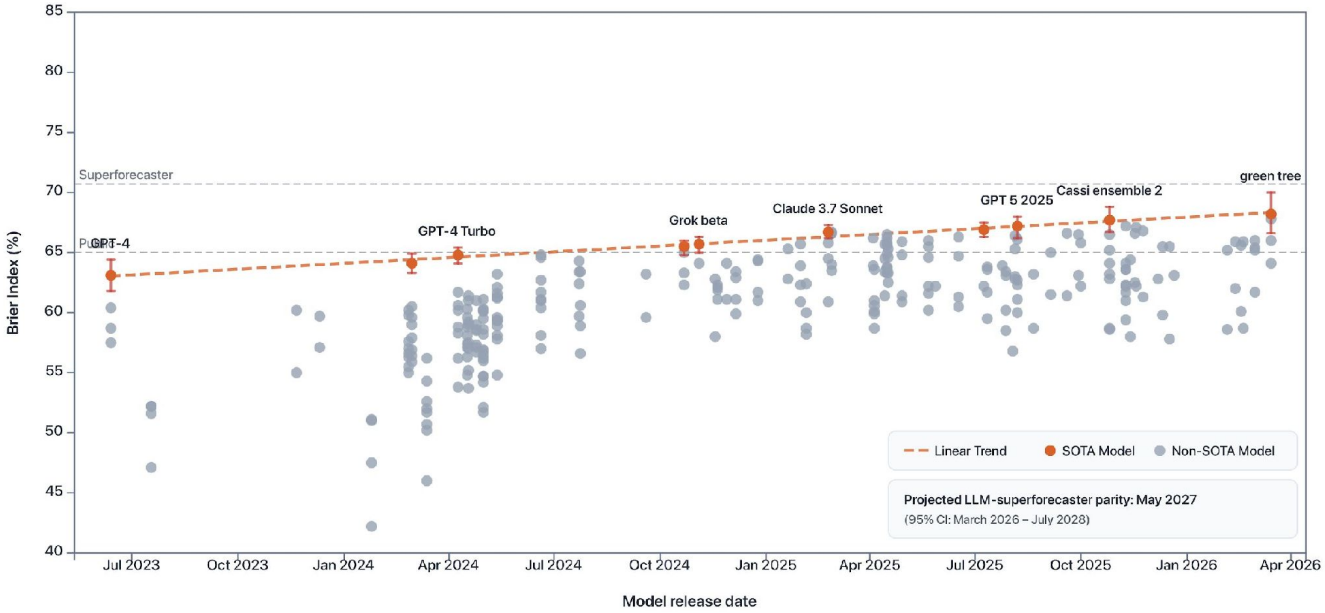
Unsaturable benchmark

The future genuinely can't be memorized or gamed.

Forecasting stress-tests everything hard about agents at once.

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AI forecasting keeps improving



ForecastBench (Karger, Tetlock et al.) · Brier Index by model release date

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But it has to earn its place.

Impressive on a benchmark isn't useful on your problem — the test is beating established methods, honestly.

> That comparison is what the bootcamp is built around.

Two questions, two days

1. **Can LLMs and agents forecast?**
   Do they act as effective time-series forecasters — measured honestly, against real baselines?

2. **How else does agentic AI apply?**
   How do advances like self-adaptation and agentic evaluation carry over to forecasting and prediction?

Why agentic forecasters?

01

Flexible

Dense series, sparse events, and text — one pipeline.

02

Scalable

One base model, many configured agents.

03

Steerable

Opinionated: priors and evidence strategies.

04

Adaptive

Learns from feedback and experience.

05

Pluggable

Interoperable across decision pipelines.

Skills + code in the Analyst Agent; self-adaptation in the Adaptive Agent.

Two ways to use a forecaster

TRACK 1 · FORECASTERS

Emits a standard Prediction

Scored head-to-head

CRPS · Brier · RPS

Agents as forecasters

TRACK 2 · ANALYSTS

Scenario analysis & Q&A

Code-backed exploration

Not head-to-head scored

Agents as prediction analysts

Same agent backbone — one config decides which.

The reference implementations

Five self-contained experiments — numerical → LLMP → agents

Use case	Scored as	What it adds
Getting started	CRPS	The minimal loop
S&P 500	CRPS	Conventional bake-off
Food prices (CFPR)	CRPS	Report-grounded LLMP
Energy / WTI	CRPS · Brier	The agentic staircase
BoC rate calls	RPS · Brier	LLM-as-judge eval

Each mixes multiple methods and ships a 99_starter_agent.ipynb.

All on one shared Predictor interface.

Getting the most out of it



Mind the compute budget

Default to **Gemini Flash Lite** — a full backtest on Gemini 3.5 can burn your budget.



Go wide, then deep

Two days can't cover it all — map what we cover to your use case first, then dig in.



Ask the concierge

Use the concierge agent for high-level questions about the repo and methods.

Reserve the advanced model for the critical path or a smaller backtest subset.

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Time Series Forecasting Foundations

Comparing classical methods, LLMs, and agents — honestly

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What we'll cover

01

Methods

A short lineage — ending at LLMs & agents.

02

Honest evaluation

The shared skeleton, plus cutoffs, CRPS & leakage.

03

A worked forecast

Real prices, real score — where numbers-only breaks.

A short history of methods

Era	Representative methods	What it added
Statistical	ARIMA, ETS, Kalman	Calibrated intervals
Machine learning	XGBoost, LightGBM	Covariates
Deep learning	DeepAR, N-BEATS, TFT	Representations
Foundation models	TimesFM, Chronos	Zero-shot transfer
LLMs & agents	LLM Processes, agents	Language + reasoning

Same question, same interface, same score.

A 50-year-old model and an LLM agent, judged the exact same way.

That's what makes a fair comparison possible.

HONEST EVALUATION

Measuring a forecaster

What makes a comparison honest

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Anatomy of a forecast eval

1. **Task**
   What to forecast: series, horizon, frequency
2. **Origin & cutoff**
   When you stand — you see only data available then
3. **Predictor**
   How you answer — the only part that varies
4. **Resolution**
   The ground truth at the target date
5. **Score**
   How close the probabilistic answer was (CRPS / Brier / RPS)

What makes a comparison honest

NUMERICAL METHODS

See only data ≤ origin

Cutoff-safe by construction

Backtest any period

LLMS / AGENTS

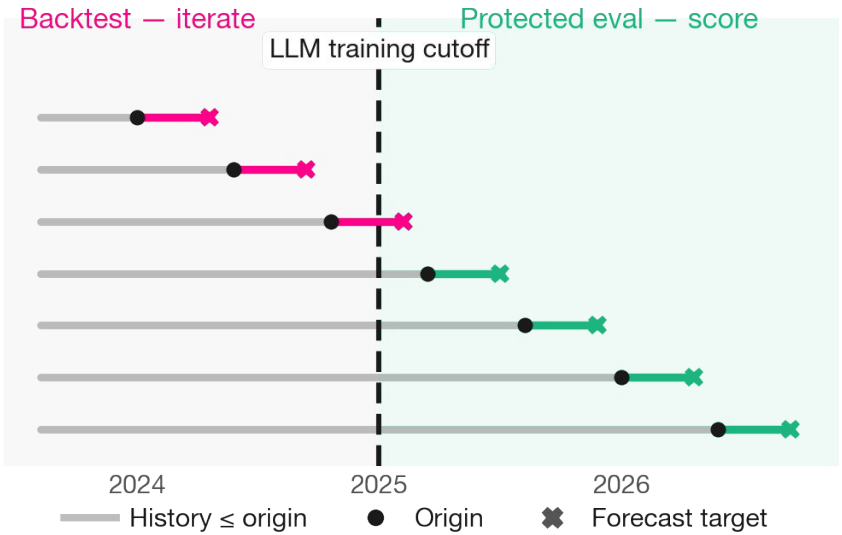
Trained up to ~Jan 2025

Pre-cutoff = recalled, not forecast

Honest only after cutoff

So we score LLM rows on a protected, post-cutoff window.

Backtest vs. protected eval



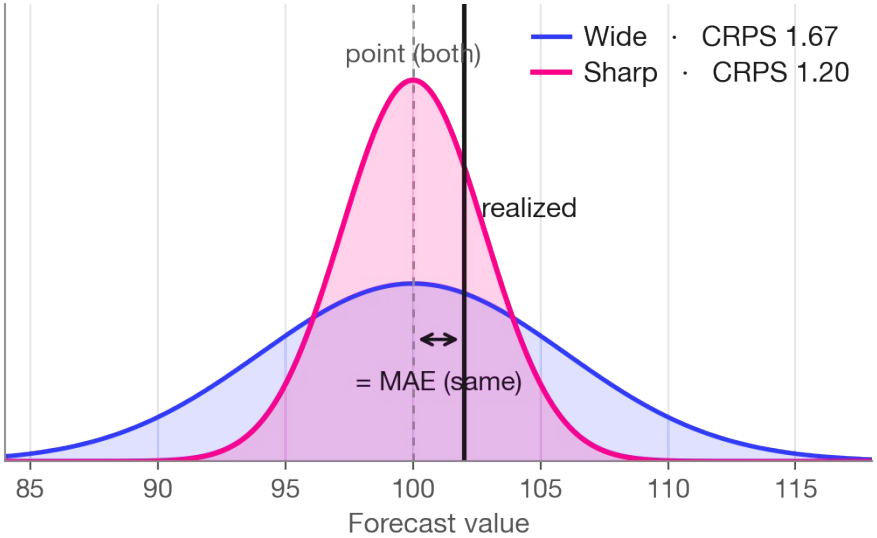
Rolling-origin eval · backtest to iterate · post-cutoff to score

The cutoff is the experiment.

Each forecast sees only its past — score on a protected post-cutoff window.

- Backtest: iterate fast
- Eval: held out, scored

What the score rewards



Two forecasts, same point forecast — CRPS separates them, MAE can't

Right place, tight interval.

CRPS scores the whole distribution, not just the point — sharper, well-placed wins.

- MAE sees only the point
- CRPS rewards sharpness

One interface, any method

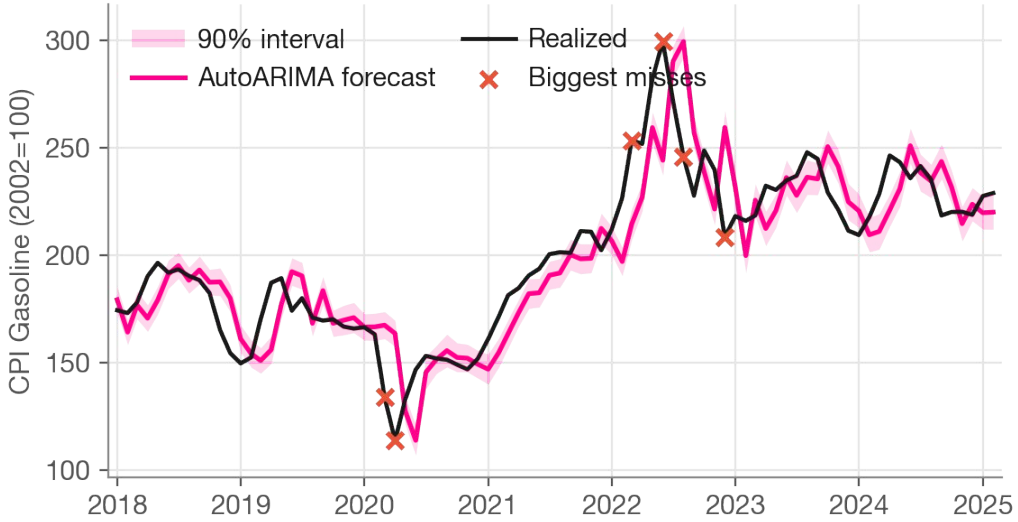
```python
class Predictor(ABC):
    @abstractmethod
    def predict(self, task, context):
        ...  # -> list[Prediction]
```

aieng/forecasting/evaluation/predictor.py

One predict() per method.

- Naive — few lines
- Darts ARIMA wrapper
- LLM-Process: prompt
- Tool agent: loop

A forecast — and where it breaks



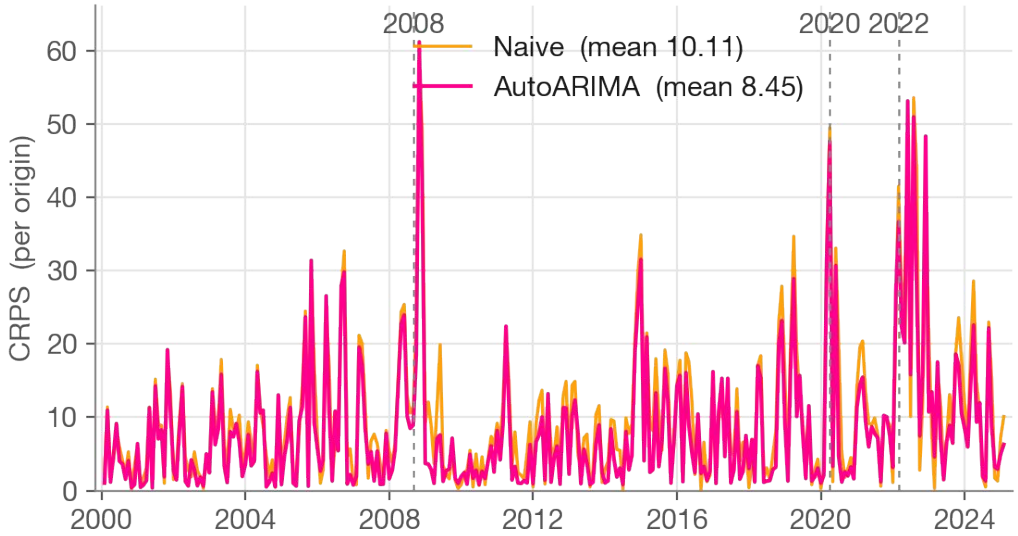
CPI Gasoline · 1-month AutoARIMA · 90% interval

It can't see the news.

AutoARIMA tracks the level but lags every turning point — it's essentially extrapolating.

- Biggest misses: 2020 crash, 2022 surge
- Both driven by news a numbers-only model never sees

The error clusters at the shocks



Per-origin CRPS · Naive vs AutoARIMA · 301 origins, 2000–2025

AutoARIMA wins — by ~16%.

Beats the floor — yet both spike at 2008, 2020, 2022, where context could help.

8.45

AutoARIMA · mean CRPS

10.11

Naive floor

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What to take forward

01

Honest evaluation, always

One interface, strict cutoffs, protected evals — for every method.

02

Established methods set the bar

A real classical model clearly beats the naive floor — tough to top.

03

Agents earn their place

By bringing context and reasoning at the shocks — tested, not assumed.

Up next: the conventional methods that set the bar — over to Behnoosh.